

Group Participation Sheet

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A portion of this course is devoted to group work. This **Group Participation Sheet** (GPS), hereinafter referred to as the **GPS Sheet**, will specifically indicate the group members that actively contributed to the assignment, lab project, deliverable, or report. All names indicated on this GPS sheet will receive the same grade for the attached group work. Group members with names and signatures not shown on this GPS sheet will receive a grade of “**zero**” for the mentioned assessment. Any concerns regarding this matter should be brought to the professor’s attention by the team leader prior to the submission deadline.

Please fill the table below electronically. Group members that participated in this evaluation should **write their names** and append their **signature**. The signed GPS sheet should be added as the **first page of all group evaluations**. This should be **applied** to all evaluations uploaded on SLATE as **softcopy**, and printed/stapled and presented as **hardcopy** to your professor.

Course Title:			
Evaluation Name:			
Team Number:	N/A		
Referencing and Citation Style:	APA Style 	IEEE Style 	N/A
Group Members:	First Name and Family Name of group members: (1) (team leader)	Signature of group members: Afolabi Adesina Digitally signed by Afolabi Adesina Date: 2026.06.17 15:53:58 -04'00'	
	(2)		
	(3)	Xian Qin Digitally signed by 6d19637b-b2a2-4dbb-9e5e-df2e0d713851 Date: 2026.06.17 12:44:27 -04'00'	
	(4)	Yanan Yang Digitally signed by Yanan Yang Date: 2026.06.16 19:19:16 -04'00'	
	(5)		
Submission Date:			

Grade: /5

AI/ML Capstone Project

PR1 to PR7 Weekly Progress Report (5% of final mark)

Due: Week 6 to 11 (softcopy on SLATE, hardcopy to the Professor in class)

Name 1: Afolabi Adesina	Student Number 1: 991808102
Name 2: Evans Frimpong	Student Number 2: 991771602
Name 3: Xian Qin	Student Number 3: 991578381
Name 4: Yanan Yang	Student Number 4: 991841037

Instructions:

- Please work on the following weekly progress report with the group.
- Prepare the deliverable report as discussed in class.
- Type the report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload the report in PDF format on SLATE. The team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stapled) of the report to the Professor in class.
- Any form of plagiarism and use of AI generated text in any part of the deliverable will result in a grade of zero for all members in the group. Teams involved in this form of academic misconduct will be reported for further disciplinary actions.

Overall Objective:

Over a 7 week period, students will engage in a sprint development cycle to deliver iterative progress with the AI/ML capstone project. Each week, the designated TL is responsible for submitting a report on behalf of the group. The dates for the working weeks are as follows:

Progress Reports	Working Week	Submission Deadline
PR1 (5%)	W5 and W6: Wed. Jun. 3, 2026 to Tue. Jun. 9, 2026	W6: Fri. Jun. 12, 2026
PR2 (5%)	W6 and W7: Wed. Jun. 10, 2026 to Tue. Jun. 16, 2026	W7: Wed. Jun. 17, 2026
PR3 (5%)	W7 and RW: Wed. Jun. 17, 2026 to Mon. Jun. 22, 2026	RW: Mon. Jun. 22, 2026
PR4 (5%)	RW and W8: Tue. Jun. 23, 2026 to Tue. Jun. 30, 2026	W8: Wed. Jul. 1, 2026
PR5 (5%)	W8 and W9: Wed. Jul. 1, 2026 to Tue. Jul. 7, 2026	W9: Wed. Jul. 8, 2026
PR6 (5%)	W9 and W10: Wed. Jul. 8, 2026 to Tue. Jul. 14, 2026	W10: Wed. Jul. 15, 2026
PR7 (5%)	W10 and W11: Wed. Jul. 15, 2026 to Tue. Jul. 21, 2026	W11: Wed. Jul. 22, 2026

* RW equals Reading Week

Instructor: Prof. Mouhamed Abdulla

The weekly PR report should address these sections. Use bulleted terms with the answers.

Q1: Summary of Work

Provide a summary of the work completed during this current sprint. [1 mark]

- Yanan: Reviewing the reference paper, preprocessing the two dataset, and setting up the training experiments.
- Evans: Reviewed application frontend and api connections needed to make the application functional.
- Afolabi: Working on fairness metrics for this project in terms of the ethics of data collection.
- Xian: Test run python code with APIs from government weather website.

Q2: List of Challenges

Provide a list of challenges encountered, and how the team addressed those challenges during this sprint. [1 mark]

- Yanan: overwhelming data infos, there're 44 features in UK dataset, and 66 features in US datasets, review and pick up the necessary features that can be used to train. And there're also challenges to encodes the features to merge them. The models used to train also need to check with paper to find top 3 used to set up the initiate experiments.
- Availability of ample image dataset for training to check for road conditions.
- Xian: Some APIs from government websites are not accessible or require certain language.

Q3: Technology Stack

Provide the current technology stack used with specific version numbers for tools, libraries, databases, and APIs. [1 mark]

- Yanan: Colabs, pandas, sklearn, sklearn.neural_network, sklearn.linear_model, sklearn.metrics.
- Flask API, YOLOv8 (image detection).
- Xian: Python, ApachePackages, pandas, opendatatoronto.
- Afolabi: VS Code, Jupyter Notebook, Cursor, PyTorch and TensorFlow libraries for neural networks.

Q4: Meeting Minutes

Provide the details of the meeting minutes (those conducted in class and outside) and the GitHub activity. [1 mark]

- Yanan: Refine the workflow, and Evans proposed that one of feature road_condition if we could not find useful dataset to train, we can only use it to calculate the safety scores. Upload the dataset (processed ones) to GitHub.
- Xian: suggested some optimization suggestions to algorithms.
- Afolabi: Suggested using a Gantt chart to list tasks and due dates for each person to manage timelines.

Q5: Future Work

Provide a work plan for the upcoming week. [1 mark]

- Yanan: checking the training results and may try the different experiments.
- Evans: Establishing a solid output for image training and determine how to factor it into the final score for the road safety recommendation.
- Afolabi: Plans to use Cursor app to manage the ML modelling.
- Xian: Keep searching for APIs from published websites to grab realtime data flow.

References (Include potential references found and used to help with the AI/ML capstone project during this sprint).

References

- [1] J. Jiang, Y. Miao, and D. Wu, “Machine Learning based Prediction Analysis of Potential Factors in Traffic Accidents,” *Applied and Computational Engineering*, vol. 99, no. 1, pp. 112 to 120, Nov. 2024. [Online]. Available: <https://doi.org/10.54254/2755-2721/99/20251788>
- [2] C. Palay, “SDOT collisions all years [Data set],” City of Seattle ArcGIS Online, 2023. [Online]. Retrieved September 6, 2024, from <https://data-seattlecitygis.opendata.arcgis.com/maps/SeattleCityGIS::sdot-collisions-all-years-2>
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- [4] Microsoft, “Visual Studio Code Docs,” 2024. [Online]. Available: <https://code.visualstudio.com/docs>
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